

Force GPS Init Removal & GPS Config Default

Date: 2026-08-19 (rolling; Supplement 2 2026-08-20) **Scope:** FC (UAVXArmQ) + GCS (uavx-python/src/) + kits + legacy emulation mirror **Status:** Implemented — builds clean (FC fc_build.py OK, GCS py_compile OK all kits)

1. Symptom reported

Pressing **Force GPS Init** in the GCS:

- USB link to the FC was **lost** (host port error), and
- the GPS module kept blinking (data still being received), but
- nothing updated on the GCS.

Diagnosis: the reset was an **independent watchdog (IWDG) trip**, not a USB fault.

2. Root cause

miscInitialiseGPS → InitGPS() ran **synchronously inside the main loop** (telem.c ProcessRxPacket → CheckTelemetry → main loop WdtFeed()). The IWDG is armed with an ~8.19 s ground reload (wdt.c:51,54, prescaler 64 @ 32 kHz LSI → 4095 steps). InitUbxGPS() blocked the main loop ≈ **9.4 s**, so WdtFeed() never ran and the watchdog reset the FC:

Step	Blocking
5 baud × 2 sends UbxInitPort (gps.c:987-990)	5 × 2 × 400 ms = 4000 ms
Delay1mS(1000) after scan (gps.c:997)	1000 ms
UbxDisableNavMessages ≈ 60 × 50 ms (gps.c:457-458,451)	~ 3000 ms
SetTimepulse/SetMode/SBAS/Interval/350ms + 2 EnableMsg	350 ms
UbxSaveConfig (gps.c:512)	1000 ms
Total	≈ 9.4 s

Boot-time InitGPS() (uavxarm-v3-gke.c:181) is immune because it runs *before* WdtInit() (:203).

3. Why removal (not a fix)

- The only viable GPS-init path on this FC is **boot-time init** or a **config-change reboot** (ParamCommit → systemReset). A live re-init therefore buys nothing a reboot doesn't already provide.
- A config change (incl. any Config bit) already forces a reboot, and the GPS bit is a **boot-time decision** — there is no live-update path by design.
- Making the init cooperative (the alternative) adds a state machine plus a protocol ACK for a feature that is redundant. **Feature removed.**

4. Changes

FC — UAVXArmQ/src/telem.c

- Enum MiscComms: slot 9 miscInitialiseGPS → miscUnusedGPSInit (reserved). Implicit numbering preserved so miscForceDefaults=10, miscSetNavMode=11, miscCycleBBLog=12 are unchanged on the wire.
- case miscInitialiseGPS: handler **deleted** (was: InitGPS();).

GCS — uavx-python/src/ (+ synced to linux|macos|windows/src/)

- ui/main_window.py: **Force GPS Init button removed** (layout + widget), the self.gps_init_btn.clicked binding and the force_gps_init() method deleted. dump_black_box() restored to class-method indentation (was briefly mis-nested during edit; re-collapsed correctly).
- protocol_enums.py: GPS_INIT = 9 removed from MiscCommand; slot 9 kept as a comment (wire-stable against the FC enum).

GPS config default — GPS now on by default

- FC params.c ParamTable: pConfig2Bits default 1 (UseBatteryCompMask) → UseBatteryCompMask | UseGPSSMask (... | 8).
- GCS ui/parameter_window.py _load_defaults: Config2Bits default eUseFastStart → eUseFastStart | eUseGPS.
- Rationale: the GPS bit is the boot-time “wait for a fix before arming” gate (F.HaveGPS = UsingGPS at gps.c:1190, eInitialisingGPS at uavxarm-v3-gke.c:335). A fitted GPS should be on by default; clearing the bit remains available for airframes that must skip the fix wait.

Legacy emulation mirror — uavx-python/UAVX/

- telem.h / telem.c mirrored the same enum-slot + handler removal for consistency (not compiled into the GCS at present).

5. Verification

- FC: python3 scripts/fc_build.py → **BUILD OK** (.bin 223596 bytes).
- GCS: python3 -m py_compile OK on main_window.py, parameter_window.py, protocol_enums.py for **canonical + all three kits** (linux/macOS/windows).
- sync_kits.sh itself uses rsync (unavailable in the Flatpak sandbox), so the three changed files were mirrored with cp.
- No residual references: grep miscInitialiseGPS|GPS_INIT|gps_init|force_gps_init → clear in FC src/, GCS canonical src/, and UAVX/ mirror.

6. Notes for the user

- Flash the freshly built **UAVXArmQ/obj/FLYINGRCF4WINGMINI/FLYINGRCF4WINGMINIQ_r0.bin** to the FC.
- Existing .af files that already carry CONFIG2_BITS keep their own value (.af value wins over defaults when loaded) — the new default only affects fresh/LoadDefaults configs.
- PDF generation is the user’s job on host “Merlin” (scripts/md2pdf.sh).

Supplement 1 — Build Target Fix + GPS ISR Refactor (2026-08-19)

7. Wrong default build target (USART1 GPS never fed)

scripts/fc_build.py hardcoded the default board to UAVXF4V3 (BOARD env default) while the Makefile defaults to FLYINGRCF4WINGMINI (Makefile:23). Consequences when flashing the default artifact:

- uavxf4v3.inc:138-148 maps TelemetrySerial = eUsart1 and GPS to eUsart2 (only when pRxType == eCPPMRx). A GPS wired to **USART1 (PA9/PA10)** thus received bytes, but SerialISR(1) never routed them to GPSISR — the GPS LED blinks, zero GPS data in the FC/GCS.
- flyingrcf4wingmini.inc:124-130 correctly maps GPS to eUsart1 and telemetry to USB VCP.

Fix: fc_build.py default BOARD → FLYINGRCF4WINGMINI (AGENTS.md FC Build section updated). Diagnosed by the user: “default build is incorrect in the Makefile” — the Makefile was already correct; only the replica was wrong.

8. GPS packet processing moved out of the ISR

GPSISR ran ProcessGPSSentence() (validity gates, cosf, GPSKFUpdate, UpdateWhere) synchronously inside the USART1 interrupt. That is heavy work in ISR context and the likely cause of the **periodic ~90% exec-time bar** (exec% = 100 * execTimeuS / CurrPIDCycleuS, telem.c:559; cycle = 2 ms).

Refactor (binding principle: **ISRs are minimal execution time — only simple state machines and flag-setting; heavy math/nav runs in the main loop**):

Before	After
GPSISR: byte FSM + ParseUbxPacket() + ProcessGPSSentence() + Kalman — all interrupt context —	GPSISR / RxGPSUbxPacket: byte FSM only ; on valid checksum set F.GPSPacketReceived (gps.c:1096-1101) UpdateInertial() (inertial.c:556-567) polls the flag each PID cycle: ParseUbxPacket() → ProcessGPSSentence() → Kalman corr./UpdateWhere, then GPSKFPropagate (predict) runs in DoSensorUpdate

- gps.h: ParseUbxPacket, ProcessGPSSentence exported.
- Emulation unaffected: emu.c:647 sets F.GPSValid directly, bypassing the ISR/parse path entirely.
- Overrun note (user-acknowledged): the shared ubx.payload union is still written by the ISR; switching to a snapshot/double-buffer is a documented follow-up if inter-packet spacing ever tightens (packets » 2 ms main-loop period today, so the flag is consumed well within packet gaps).
- **Follow-up (accepted, deferred):** close the acknowledged overrun window by either (1) double-buffering the packet in the ISR (flip an index on completion — ISR stays byte-FSM only, main decodes its own buffer), or
(2) snapshot memcpy (≤384 B) of ubx.payload in the main loop at flag time before decode. Design agreed: ISR = byte FSM + flag only; decode (ParseUbxPacket) + KF + Nav run in UpdateInertial. Do not decode into GPS.* from the ISR — it recreates the same race downstream.

- `F.NewNavUpdate` (GPS fix cadence) and `F.GPSValid` now update at the PID rate from main-loop context, same 5-10 Hz data cadence, feeding `auto.c:439` as before.

9. AGENTS.md

Binding rule added under **FC Style Guide**: “ISRs must be minimal execution time — only simple state machines and flag-setting/queue-feeding; never run Kalman filters, trig/fp math, navigation, or EEPROM/flash writes from interrupt context.”

10. Verification

- FC: `python3 scripts/fc_build.py` → **BUILD OK**, `FLYINGRCF4WINGMINIQ_r0.bin` 222444 bytes (this session, at each refactor step).
- Default-board check: `fc_build.py` reports `default BOARD = FLYINGRCF4WINGMINI`.

11. Remaining investigation (user)

With the correct target flashed: “Exec time bar periodic ~90%, some acc misreads (acc cal done once), GPS LED blinking but nothing displayed.” IMU read path (`ReadIMUAccAndRate`, `icm426xx.c:84`) busy-waits on an `INT_STATUS` data-ready edge with a **5000 µs deadline** — a slow/missing `DRDY` edge could block `UpdateInertial` and inflate the exec bar; and GPS still shows nothing until the UBX frame actually parses (fail-safe to check: GPS module emitting NMEA, not UBX, after init misses — parser is UBX-only). —

Supplement 2 — GPS Pass-Through Removed + USART RX Macro Bug Found (2026-08-20)

12. GPS pass-through fully removed

The GCS “GPS Pass-Thru” feature (added 2026-08-19 to relay raw GPS bytes to the telemetry link when GPS parsing showed nothing) was **removed entirely** — FC and GCS — at the user’s request (“the pass thru code can go away and the selector marked as unused”). Raw GPS sniffing is not a viable diagnostic path: if the FC parser shows nothing, routing bytes to the host adds little without u-center, and “if we have to resort to ucenter we would be in terrible trouble” (user).

Side	File	Change
FC	<code>telem.c</code>	<code>miscGPSPassThru</code> → <code>miscUnusedGPSPassThru</code> (slot 5 kept, wire-stable); dispatch case deleted; globals <code>EnableGPSPassThru/GPSPassThruSerial</code> removed
FC	<code>sensors/gps.c</code>	<code>CheckGPSUpdate()</code> restored to plain UBX poll/parse + timeout (pass-thru bridging branch deleted)
FC	<code>telem.h</code>	extern decls for the two globals removed
GCS	<code>protocol_enums.py</code>	<code>GPS_PASS_THRU = 5</code> → <code>GPS_PASS_THRU_UNUSED = 5</code>

Side	File	Change
GCS	ui/parameter_window.py	“GPS Pass-Thru” button + toggle_gps_pass_thru() removed
GCS	ui/calibration_window.py	enum in misc-name map renamed to match

Cost of removal: none. The feature was never relied on in flight; UBX parsing is unchanged. The selector byte is preserved so a stale GCS (or a recorded session) cannot accidentally re-enter a removed mode.

13. Root cause of “GPS receives nothing”: same-port USART RX pad never set to AF

Symptom (UAVXF4V3, no USB): GPS LED blinking but no GPS data; also sporadic telemetry command dropouts — consistent with dead USART **receive**.

Root cause — InitSerialPort() (boards/harness.c, all targets), **GPIO block**:

	Tx pin	Rx pin
Legacy (Arm32F4)	GPIO_Pin = Tx.Pin Rx.Pin — one	same call
Current fork (before fix)	GPIO_Init, both pads → GPIO_Mode_AF GPIO_Pin = u->Tx.Pin → AF	RX pad only re-initialised if Rx.Port != Tx.Port (harness.c:585)

For every same-port USART on the hardware (USART1 PA9/PA10, USART2 PA2/PA3, USART3 PB10/PB11, UART4 PC10/PC11), the RX pad was left in whatever the bulk init left it — GPIO_Mode_AN (analog, harness.c:857) — and GPIO_PinAFConfig() only attaches the AF *function*; an AF pad that is still in analog mode is **disconnected from the USART receiver**. RXNE never fires → no bytes in RxQ → GPS (and every same-port UART receive) dead.

This is the “USART/UART initialisation break” suspected after the USB-signalling harness rework: the cross-port if guard was added to keep the RX branch from re-touching the TX port, but it skipped the same-port case entirely.

Fix (boards/harness.c): unconditionally re-init the RX pad on its own port:

```
GPIO_StructInit(&GPIO_InitStructure);
GPIO_InitStructure.GPIO_Speed = GPIO_Speed_50MHz;
GPIO_InitStructure.GPIO_Pin = u->Tx.Pin;
GPIO_InitStructure.GPIO_Mode = GPIO_Mode_AF;
GPIO_Init(u->Tx.Port, &GPIO_InitStructure);

/* The RX pin must be set to AF mode on its own port regardless of */
/* whether it shares the TX port ... */
GPIO_InitStructure.GPIO_Pin = u->Rx.Pin;
GPIO_Init(u->Rx.Port, &GPIO_InitStructure);

GPIO_PinAFConfig(u->Tx.Port, u->Tx.PinSource, u->USART_AF);
GPIO_PinAFConfig(u->Rx.Port, u->Rx.PinSource, u->USART_AF);
```

13.1 Blast radius (USB targets — safe)

The RX-pad init runs inside `if ((s > eUSBSerial) && (s < eSoftSerial))` (harness.c:576) — **hardware UARTs only**. `eUSBSerial` (USB VCP, `TM_USB_VCP_Init`, `OTG_FS` clock, PA11/12 D+/D-) and `eSoftSerial` are excluded, so **USB comms on every target are untouched**. The single shared-pad case — `flyingrcf4wingmini.inc:31 CPPMPin = PA3` (TIM2 CH4, == USART2 RX pad) — is safe by call order: `InitSerialPort()` (during `InitTarget`, harness.c:879) then `InitRC()` → `InitCPPMPin()` (harness.c:913) re-maps PA3 to TIM2 AF, last write wins. Net behaviour change everywhere: “hardware UART RX pads are correctly AF-mapped”, a strict restoration of the legacy init.

14. UAVXF4V3 GPS path verified end-to-end

- Target maps `TelemetrySerial = eUsart1` (PA9/PA10), GPS to `eUsart2` (PA2/PA3) when `pRxType == eCPPMRx` (default); CPPM capture uses **PA0** so it does not conflict with USART2 (the `rc.c` comment “PA3 remuxed TIM2 CH4” is a stale cross-board comment for this target; `uavxf4v3.inc:27` CPPM is PA0).
- `InitGPS()` (`uavxarm-v3-gke.c:183`) runs unconditionally before `WdtInit()`: `SetBaudRate 115200` → `InitUbxGPS` (iNav baud-scan + UBX nav5/GNSS/interval config). **Parser is UBX-only** (`RxGPSUbxPacket`) — an NMEA-only module after a missed init produces silence; a u-blox M8/M9/M10 is required.
- **Build:** `BOARD=UAVXF4V3 python3 scripts/fc_build.py` → **BUILD OK**, `UAVXF4V3Q_r0.bin` 225396 bytes.

15. Verification

- `FC grep GPSPassThru` → only the retained enum-rename comment in `telem.c`.
 - `GCS grep GPS_PASS_THRU` → only `GPS_PASS_THRU_UNUSED` in `protocol_enums.py`.
 - `py_compile OK`: `parameter_window.py`, `calibration_window.py`, `protocol_enums.py`; FC build OK (above). (SVN commits done by the user on host; no `svn` binary in the Flatpak sandbox.)
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